

Module 4: Non-Deterministic Finite Automata & Subset Construction

Comprehensive Identification & Multiple-Choice Reviewer — 45 Items

Name: _____

Date: _____ Score: _____

INSTRUCTIONS: Read each statement carefully. Identify the correct term or concept described in the blank (____). Choose the letter of the correct answer from the choices provided (a, b, c, or d) and write it on the blank provided. An exhaustive Answer Key with detailed rationales is provided at the end of this document.

PART I: IDENTIFICATION QUESTIONNAIRE (45 ITEMS)

- ____ 1. A state machine where a single input symbol can lead to multiple next states or no next state at all is a ____ Finite Automaton.
a.) Linear Bounded b.) Deterministic c.) Non-Deterministic d.) Turing
- ____ 2. Unlike a DFA, an NFA is permitted to make transitions without consuming any input symbol, which are known as ____-moves.
a.) Q b.) ϵ (Epsilon) c.) δ (Delta) d.) Σ (Sigma)
- ____ 3. For an NFA, the transition function δ maps each state and input symbol (including ϵ) to ____.
a.) exactly one single state (Q) b.) the alphabet (Σ) c.) a subset of states (2^Q or power set) d.) a stack of tokens
- ____ 4. If for a given state q and symbol a there is no valid transition in an NFA, the transition maps to the ____ set (ϕ).
a.) empty b.) final c.) universal d.) initial
- ____ 5. A computational branch in an NFA that reaches the empty set (ϕ) with no available transitions is known as a dead ____.
a.) tape b.) stack c.) configuration d.) symbol
- ____ 6. In terms of language recognition capability, how does the expressive computational power of an NFA compare to that of a DFA?
a.) NFA recognizes Context-Free languages b.) DFA is strictly more powerful c.) They are exactly equal (both recognize Regular Languages) d.) NFA is strictly more powerful
- ____ 7. While DFAs and NFAs recognize the exact same class of languages, NFAs are often much easier to design and can have ____ states than equivalent DFAs.
a.) fractionally more b.) infinitely more c.) exactly double d.) exponentially fewer
- ____ 8. In a DFA, how many next states must exist for every (state, input symbol) pair?
a.) Exactly one b.) Zero or more c.) At least two d.) Any arbitrary subset
- ____ 9. According to the formal acceptance rule, an NFA accepts an input string if ____.
a.) the machine has no dead configurations b.) at least one computational branch ends in an accept/final state c.) all computational branches end in an accept state d.) the majority of branches accept
- ____ 10. An NFA rejects an input string if and only if ____.
a.) the string contains an odd number of zeros b.) at least one path fails c.) ALL computational paths fail to reach any final state d.) a dead configuration is encountered on one path
- ____ 11. In the formal 5-tuple $M = (Q, \Sigma, q_0, F, \delta)$ of an NFA, the power set of states Q is mathematically represented as ____.
a.) Q^2 b.) Σ^Q c.) 2^Q d.) $Q!$
- ____ 12. If an NFA has 3 states, how many elements are in its power set 2^Q ?

a.) 9 b.) 3 c.) 6 d.) $8(2^3)$

_____ 13. In Week 4 Example 1 (NFA for strings ending with '0'), state A loops on {0, 1} and transitions on '0' to state B. When input '100' is processed, what is the set of states occupied at the end?

a.) $\emptyset$ b.) {A, B} c.) {A} d.) {B}

_____ 14. In Week 4 Example 1, why is the input string '100' accepted by the NFA?

a.) Because the string has 3 symbols b.) Because state A is an accepting state c.) Because state B is in {A, B} and B is an accepting state d.) Because there are no 1s at the end

_____ 15. In Week 4 Example 2 (NFA for strings starting with '0'), what occurs if the first input symbol is '1'?

a.) The machine complements the string b.) The machine enters a dead configuration ($\emptyset$) and rejects c.) The machine loops at state A d.) The machine transitions to state B

_____ 16. In Week 4 Example 3, an NFA accepts all strings over {0, 1} of length exactly 2 with states A $\rightarrow$ B $\rightarrow$ C. Why is string '01' accepted?

a.) Because A is an accepting state b.) A transitions on 0 to B, and B transitions on 1 to final state C c.) Because C loops on all inputs d.) Because it has no trap states

_____ 17. The algorithmic method used to convert any Non-Deterministic Finite Automaton (NFA) into an equivalent Deterministic Finite Automaton (DFA) is known as the _____ construction.

a.) Chomsky b.) Turing c.) pumping d.) subset (powerset)

_____ 18. In the subset construction algorithm, each individual state in the resulting DFA represents a _____ of states from the original NFA.

a.) permutation b.) quotient c.) subset (or set) d.) derivative

_____ 19. If an NFA has n states, what is the theoretical maximum number of states the converted DFA could possess before minimization?

a.) n^2 b.) $n!$ c.) 2^n d.) $2n$

_____ 20. During subset construction, how is the transition on symbol 'a' from a composite DFA state $S = \{q_1, q_2\}$ computed?

a.) By taking the union of transitions: $\delta(q_1, a) \cup \delta(q_2, a)$ b.) By taking the Cartesian product c.) By choosing only the transition from q_1 d.) By taking the intersection: $\delta(q_1, a) \cap \delta(q_2, a)$

_____ 21. In subset construction, which subsets of NFA states are designated as final (accepting) states in the converted DFA?

a.) Only subsets containing exclusively final states b.) Any subset that contains at least one final state of the NFA c.) Only the initial subset d.) Subsets containing the empty set

_____ 22. In subset construction, if a state transition leads to the empty set ($\emptyset$), how is this represented in the equivalent DFA?

a.) It is designated as the start state b.) It becomes an accepting state c.) It is completely removed from the DFA d.) As a dead (trap) state that loops on all input symbols

_____ 23. According to the set algebra rule cited in Slide 10 of Week 4, what is the result of taking the union of set A with the empty set $\emptyset$ ($A \cup \emptyset$)?

a.) A b.) $\cup$ c.) $\emptyset$ d.) {A, $\emptyset$ }

_____ 24. In Slide 10 (NFA to DFA conversion for strings ending with '1'), the NFA has states {A, B} where A is start and B is final. What are the states in the converted DFA?

a.) State A and State B b.) Only State A c.) State A and State AB (where AB is accepting) d.) State A, State B, State C

_____ 25. In Slide 10, what is the transition from DFA state AB on input '0'?

a.) It transitions back to state A b.) It transitions to state B c.) It transitions to a dead state d.) It loops on state AB

_____ 26. In Slide 10, what is the transition from DFA state AB on input '1'?

a.) It transitions to state A b.) It loops on state AB c.) It halts d.) It enters a trap state

_____ 27. In Activity 3 Problem 1 ($L_1 = \{\text{strings containing '01'}\}$), what is the regular expression for the language?

a.) $(0 + 1)^* 01$ b.) $(01)^*$ c.) $01 (0 + 1)^*$ d.) $(0 + 1)^* 01 (0 + 1)^*$

_____ 28. In Activity 3 Problem 1 ($L_1 = \{\text{strings containing '01'}\}$), how many states does the minimal DFA have?

a.) 2 states b.) 5 states c.) 4 states d.) 3 states

_____ 29. In Activity 3 Problem 2 ($L_2 = \{\text{strings starting with '10'}\}$), what is the regular expression?

a.) $1 (0 + 1)^* 0$ b.) $10 (0 + 1)^*$ c.) $(10)^*$ d.) $(0 + 1)^* 10$

_____ 30. In Activity 3 Problem 2 ($L_2 = \{\text{strings starting with '10'}\}$), why does the converted DFA require a Trap state?

a.) To permanently reject any string whose prefix is not '10' b.) To store stack memory c.) To loop on final states
d.) To handle epsilon transitions

_____ 31. In Activity 3 Problem 4 ($L_4 = \{\text{strings ending with '1'}\}$), what is the regular expression?

a.) $(1)^*$ b.) $1 (0 + 1)^*$ c.) $(0 + 1)^* 1$ d.) $(01)^*$

_____ 32. In Activity 3 Problem 5 ($L_5 = \{\text{strings ending with '11'}\}$), how many states does the equivalent minimal DFA contain?

a.) 2 states b.) 4 states c.) 5 states d.) 3 states

_____ 33. In Activity 3 Problem 3 ($L_3 = \{\text{strings containing '0'}\}$), what is the minimal number of states in the equivalent DFA?

a.) 2 states b.) 4 states c.) 3 states d.) 1 state

_____ 34. In an ϵ -NFA, the set of all states that can be reached from a state q using only empty transitions (ϵ -moves) without consuming any input is called the _____ of q .

a.) ϵ -closure (Epsilon closure) b.) Kleene closure c.) power set d.) transition matrix

_____ 35. The ϵ -closure of any state q always contains at least _____.

a.) the final state b.) the state q itself c.) the initial state d.) the empty set

_____ 36. In Activity 3 Problem 1 ($L_1 = \{\text{strings containing '01'}\}$), the DFA created by subset construction has 4 reachable states before minimization: $A = \{q_0\}$, $B = \{q_0, q_1\}$, $C = \{q_0, q_2\}$, and $D = \{q_0, q_1, q_2\}$. Which states are accepting?

a.) Only State C b.) Only State D c.) States A and B d.) States C and D (both contain q_2)

_____ 37. In Activity 3 Problem 1, why can states $C = \{q_0, q_2\}$ and $D = \{q_0, q_1, q_2\}$ be merged during DFA minimization?

a.) Because both have dead configurations b.) Both states transition to D on '0' and to C on '1', making them equivalent
c.) Because state D is unreachable d.) Because both are initial states

_____ 38. In subset construction, states in the power set that cannot be reached by any sequence of transitions starting from the initial state are called _____ states.

a.) unreachable (inaccessible) b.) trap c.) final d.) deterministic

_____ 39. The famous algorithmic technique that converts any Regular Expression into an equivalent ϵ -NFA is known as _____ Construction.

a.) Thompson's b.) Chomsky's c.) Turing's d.) Moore's

_____ 40. In an NFA, when multiple transitions exist for the same input symbol from a single state, how does the machine conceptually process them?

a.) It always chooses the alphabetically first state b.) It crashes with a runtime error c.) It prompts the user for a choice
d.) It explores all possible execution paths simultaneously in parallel

_____ 41. An NFA with n states that recognizes a language can always be converted into an equivalent DFA with at most _____ states.

a.) 2^n b.) $n!$ c.) $2n$ d.) n^2

_____ 42. In an NFA transition table, an entry with the symbol ϕ indicates that _____.

a.) no transition is defined for that state and input symbol b.) an epsilon transition occurs c.) the state is an accepting state
d.) the machine loops infinitely

_____ 43. In Activity 3 Problem 5 (strings ending with '11'), what is the transition from DFA state $B = \{q_0, q_1\}$ on input '1'?

a.) State A = $\{q_0\}$ b.) State B = $\{q_0, q_1\}$ c.) Trap state d.) State C = $\{q_0, q_1, q_2\}$ (Accepting)

_____ 44. In Activity 3 Problem 5 (strings ending with '11'), what is the transition from DFA state $C = \{q_0, q_1, q_2\}$ on input '0'?

- a.) State $A = \{q_0\}$ b.) State $B = \{q_0, q_1\}$ c.) State $C = \{q_0, q_1, q_2\}$ d.) Trap state

_____ 45. Which of the following computational characteristics is shared by BOTH DFAs and NFAs?

- a.) Both require unique deterministic transitions b.) Both require exactly one next state per symbol c.) Both recognize exactly the class of Regular Languages d.) Both allow epsilon transitions

COMPLETE ANSWER KEY & EXPLANATIONS

Module 4: Non-Deterministic Finite Automata & Subset Construction — Reference Key with Theoretical Rationales

Item	Key	Correct Answer	Theoretical Explanation & Concept
1	C	Non-Deterministic	[NFA Core Definition] A Non-Deterministic Finite Automaton (NFA) allows zero, one, or multiple outgoing transitions from a state on a single input symbol.
2	B	ϵ (Epsilon)	[Epsilon Moves in NFA] NFAs can incorporate ϵ -transitions (empty moves), enabling state changes without consuming input symbols.
3	C	a subset of states (2^Q or power set)	[NFA Transition Function Signature] The NFA transition function is defined as $\delta: Q \times (\Sigma \cup \{\epsilon\}) \rightarrow 2^Q$, meaning its output is a set of states (element of the power set).
4	A	empty	[Empty Set Transition] In an NFA, having no next state for an input symbol means $\delta(q, a) = \emptyset$, representing a dead computational branch.
5	C	configuration	[Dead Configuration] When an NFA computational path encounters a state with no transition for the current input symbol, that branch enters a dead configuration and terminates.
6	C	They are exactly equal (both recognize Regular Languages)	[DFA vs NFA Equivalence] By the Rabin-Scott powerset construction theorem, any language recognized by an NFA can also be recognized by an equivalent DFA.
7	D	exponentially fewer	[State Conciseness of NFA] An NFA with n states may require up to 2^n states when converted into a minimal equivalent DFA.
8	A	Exactly one	[DFA Uniqueness Constraint] A DFA requires exactly one deterministic next state for every combination of state and input symbol.
9	B	at least one computational branch ends in an accept/final state	[NFA Acceptance Criterion] An NFA accepts if there exists at least one valid sequence of transitions that consumes the entire string and terminates in a final state F .
10	C	ALL computational paths fail to reach any final state	[NFA Rejection Rule] Rejection requires that none of the possible parallel computational branches terminate in an accepting state.
11	C	2^Q	[Power Set Notation] 2^Q denotes the set of all subsets of Q (the power set), which is the co-domain of the NFA transition function.
12	D	8 (2^3)	[Power Set Size] The power set of a set with $ Q = 3$ contains $2^3 = 8$ possible subsets.
13	B	{A, B}	[Example 1 Trace '100'] After '100', branch A remains at A, and the '0' transition also moves to B, resulting in active state subset {A, B}.
14	C	Because state B is in {A, B}	[Example 1 Acceptance Rationale] Since state B is a final state and B

		and B is an accepting state	$\in \{A, B\}$, at least one path ends in a final state, so the string is accepted.
15	B	The machine enters a dead configuration (ϕ) and rejects	[Example 2 Dead Configuration on '1'] State A has no transition defined for '1', resulting in $\delta(A, 1) = \emptyset$ (dead configuration).
16	B	A transitions on 0 to B, and B transitions on 1 to final state C	[Example 3 Length 2 Trace] The sequence $A \xrightarrow{0} B \xrightarrow{1} C$ lands in final state C after 2 symbols.
17	D	subset (powerset)	[Subset Construction Name] The subset construction (or powerset construction) algorithm systematically models sets of NFA states as individual DFA states.
18	C	subset (or set)	[DFA States as NFA Subsets] A state in the converted DFA corresponds to a subset of NFA states that could be reached simultaneously.
19	C	2^n	[Maximum DFA State Bound] Since there are 2^n subsets of a set of n states, the converted DFA has at most 2^n states.
20	A	By taking the union of transitions: $\delta(q_1, a) \cup \delta(q_2, a)$	[Subset Transition Union Rule] The DFA transition on symbol 'a' from set S is the union of all NFA transitions from each state in S on symbol 'a'.
21	B	Any subset that contains at least one final state of the NFA	[DFA Accepting State Condition] A DFA state S is an accepting state if $S \cap F_{NFA} \neq \emptyset$ (it contains at least one state that was an accepting state in the NFA).
22	D	As a dead (trap) state that loops on all input symbols	[Dead State Representation in DFA] The empty set ϕ becomes an explicit trap state in the DFA with self-loops on all alphabet symbols.
23	A	A	[Set Union with Empty Set] Union with the empty set is identity: $A \cup \emptyset = A$.
24	C	State A and State AB (where AB is accepting)	[Slide 10 Converted DFA States] The reachable subsets are $\{A\}$ and $\{A, B\}$, where $\{A, B\}$ is accepting because it contains NFA final state B.
25	A	It transitions back to state A	[Slide 10 Transition on 0] From AB, on input 0: $\delta(A, 0) \cup \delta(B, 0) = A \cup \emptyset = A$. Thus it returns to state A.
26	B	It loops on state AB	[Slide 10 Transition on 1] From AB, on input 1: $\delta(A, 1) \cup \delta(B, 1) = \{A, B\} \cup \emptyset = \{A, B\} = AB$, looping on itself.
27	D	$(0 + 1)^* 01 (0 + 1)^*$	[Regex for Containing '01'] Strings containing '01' anywhere have the regex $(0 + 1)^* 01 (0 + 1)^*$.
28	D	3 states	[Minimal DFA for Containing '01'] The minimal DFA requires 3 states: S0 (not seen 0), S1 (seen 0), and S2 (seen 01, accepting).
29	B	$10 (0 + 1)^*$	[Regex for Starting with '10'] Strings that start with '10' followed by any sequence of symbols are given by $10 (0 + 1)^*$.

30	A	To permanently reject any string whose prefix is not '10'	[Trap State in Starting with '10'] Any string starting with '0' or '11' violates the required prefix '10' and must be diverted to an inescapable trap state.
31	C	$(0 + 1)^* 1$	[Regex for Ending with '1'] Any string ending with '1' is matched by $(0 + 1)^* 1$.
32	D	3 states	[DFA for Ending with '11'] Matching strings ending with '11' requires 3 DFA states: S0 (no trailing 1), S1 (one trailing 1), and S2 (two or more trailing 1s, accepting).
33	A	2 states	[DFA for Containing '0'] Only 2 states are needed: state A (seen only 1s, non-accepting) and state B (seen at least one 0, accepting).
34	A	ϵ -closure (Epsilon closure)	[Epsilon Closure Definition] The ϵ -closure(q) is the set of all states reachable from state q by following zero or more ϵ -transitions.
35	B	the state q itself	[Epsilon Closure Self-Inclusion] Because a state is reachable from itself via zero ϵ -moves (path of length 0), $q \in \epsilon$ -closure(q) always.
36	D	States C and D (both contain q2)	[Activity 3 Problem 1 Accepting States] Because q2 is the accepting state in the NFA, any subset containing q2 (namely C and D) is an accepting state in the DFA.
37	B	Both states transition to D on '0' and to C on '1', making them equivalent	[State Equivalence in Minimization] States C and D have identical transition outputs on all alphabet symbols ($0 \rightarrow D$, $1 \rightarrow C$) and are both accepting, proving they are indistinguishable/equivalent.
38	A	unreachable (inaccessible)	[Unreachable States in Subset Construction] While 2^Q possible subsets exist, typically only a small fraction are reachable from the start state, and unreachable subsets are discarded.
39	A	Thompson's	[Thompson's Construction] Thompson's construction algorithm systematically converts regular expressions into ϵ -NFAs using base cases and structural induction.
40	D	It explores all possible execution paths simultaneously in parallel	[Parallel Branching Concept] Non-determinism can be conceptualized as cloning the machine at each branch point to explore all computational trajectories in parallel.
41	A	2^n	[NFA to DFA State Upper Bound] By subset construction, the maximum number of states in the equivalent DFA is 2^n .
42	A	no transition is defined for that state and input symbol	[Phi Symbol in Transition Table] The symbol ϕ (or $\emptyset$) in a transition table denotes the empty set, indicating no valid next state exists for that input.
43	D	State C = {q0, q1, q2} (Accepting)	[Activity 3 Problem 5 Transition on '1'] From B = {q0, q1}, on input 1: $\delta(q0, 1) = \{q0, q1\}$ and $\delta(q1, 1) = \{q2\}$. The union is {q0, q1, q2} = C (accepting).
44	A	State A = {q0}	[Activity 3 Problem 5 Transition on '0'] From C, on input 0: $\delta(q0, 0) =$

$\{q_0\}$, $\delta(q_1, 0) = \emptyset$, and $\delta(q_2, 0) = \emptyset$. The union is $\{q_0\}$ = State A.

45

C

Both recognize exactly the class of Regular Languages

[Shared DFA and NFA Characteristics] DFAs and NFAs are computationally equivalent in language recognition power—both define and recognize Regular Languages.
